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# Early Development of Comorbidity Between Symptoms of Attention-Deficit/Hyperactivity Disorder (ADHD) and Oppositional Defiant Disorder (ODD)

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Attention-deficit/hyperactivity disorder (ADHD) and oppositional defiant disorder (ODD) are among the most common childhood disorders and frequently co-occur. The present study sought to advance our understanding of how comorbidity between ADHD and ODD develops during the preschool years by testing a cross-lagged model that integrates 2 prominent models: the developmental precursor model and the correlated risk factors model. Participants were 199 children (107 boys) who took part in a longitudinal study of preschoolers with behavior problems. Parent reports of ADHD and ODD symptoms were collected annually from ages 3 to 6 and a family history interview was administered at age 3. In support of the developmental precursors model, ADHD symptoms predicted later argumentative/defiant symptoms. In support of the correlated risk factors model, family histories of ADHD and ODD/CD symptoms were correlated risk factors that uniquely predicted ADHD and anger/irritable symptoms in children. Results suggest that the correlated risk factors model may best explain the development of comorbidity between symptoms of ADHD and anger/irritability, whereas the developmental precursors model may better explain the development of comorbidity between symptoms of ADHD and argumentative/defiance.

***General Scientific Summary***

This study indicates that there may be different reasons why symptoms of attention-deficit/hyperactivity disorder (ADHD) and oppositional defiant disorder (ODD) often co-occur. Results suggest that early symptoms of ADHD place children at risk for developing argumentative/defiant behaviors, and that ADHD and anger/irritability tend to co-occur because family histories of ADHD and ODD tend to co-occur.

*Keywords:* preschool, ADHD, oppositional defiant disorder, longitudinal

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Attention-deficit/hyperactivity disorder (ADHD) and oppositional defiant disorder (ODD) are among the most common childhood disorders, each affecting ~10% of children (Bloom, Jones, & Freeman, 2013; Nock, Kazdin, Hiripi, & Kessler, 2007). ADHD is characterized by developmentally deviant and impairing levels of inattention and hyperactivity/impulsivity, and ODD involves difficulties with anger/irritability, argumentativeness/defiance, and vindictiveness (American Psychiatric Association, 2013). ADHD and ODD alone can cause significant impairment in family, school, and social functioning, and prognosis is even worse when these disorders co-occur (Waschbusch, 2002). Comorbidity between ADHD and ODD is substantial; one-third to one-half of children with one disorder also meet criteria for the other (Nock et al., 2007; Waschbusch, 2002). Although the consequences and correlates of comorbid ADHD/ODD have been well documented (Larson, Russ, Kahn, & Halfon, 2011; Waschbusch, 2002), less well understood is why such high overlap exists.

Growing evidence that ADHD and ODD often emerge during the preschool years (Lavigne, Lebailly, Hopkins, Gouze, & Binns, 2009; Riddle et al., 2013) suggests that our understanding of the development of comorbid ADHD/ODD may be advanced by studying these symptoms early in development. High activity level, impulsive behavior, defiance, and aggression are normative during the preschool years (Campbell, 2002), but an estimated 75% to 90% of preschool-aged children who exhibit clinically significant ADHD symptoms will meet criteria for ADHD when they reach school age (Lahey et al., 2004; Riddle et al., 2013). Moreover, comorbidity between symptoms of ADHD and ODD is evident even during the early preschool years (Harvey, Friedman-Weieneth, Goldstein, & Sherman, 2007). Examining covariation between ADHD and ODD symptoms as they are emerging among preschool children with behavior problems may provide an important testing ground for theoretical models of comorbidity between ADHD and ODD.

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## Theoretical Models of Comorbidity Between ADHD and ODD

A number of models have been proposed to explain high comorbidity between ADHD and disruptive behavior disorders, the two most prominent of which are the *correlated risk factors* model and the *developmental precursor* model. The correlated risk factors model suggests that comorbidity may be a result of correlated or shared risk factors (Rhee, Willcutt, Hartman, Pennington, & DeFries, 2008). Twin studies have supported this model (Rhee et al., 2008) and generally suggest that comorbidity is because of shared genetic factors (Eaves et al., 2000; Tuvblad, Zheng, Raine, & Baker, 2009), though some have pointed to shared environmental factors (Burt, Krueger, McGue, & Iacono, 2001). The developmental precursor model posits that symptoms of ADHD lead to the development of ODD. Symptoms of ADHD in children are theorized to place stress on the family and disrupt family functioning, which in turn places children at risk for ODD (Barkley, 2006; Beauchaine, Hinshaw, & Pang, 2010; Johnston & Jassy, 2007). In support of this model, families of children with ADHD have been documented to engage in more negative parenting practices, have higher rates of parent psychopathology, and experience more parenting stress, particularly among children with comorbid ADHD and conduct problems (Johnston & Mash, 2001). Moreover, negative parenting practices have been found to mediate the relation between ADHD symptoms and conduct problems (Kaiser, McBurnett, & Pfiffner, 2011). ADHD may also lead to the development of ODD through peer influences. Children with ADHD are more likely to be rejected by peers (e.g., Hoza et al., 2005), which in turn may contribute to the development of conduct problems (Miller-Johnson, Coie, Maumary-Gremaud, & Bierman, 2002).

The early preschool years may be an optimal time to evaluate the correlated risk factors model, because the effects of such risk factors can be examined before other risk factors have exerted their influences, but after one can begin to distinguish ADHD and ODD symptoms from normative toddler behavior. The preschool years may also be a critical time for the unfolding of the developmental precursors model, because the putative mechanisms through which ADHD leads to the development of ODD may be particularly salient during this stage of development. Even in the best of circumstances, parenting preschoolers can be challenging, because noncompliance, defiance, and tantrums are normative behaviors during the preschool years (Campbell, 2002). Parenting hyperactive preschoolers may be even more stressful; the disruptions in family functioning that have been documented with older children have also been found in families of preschoolers with symptoms of ADHD (Goldstein, Harvey, & Friedman-Weieneth, 2007; Goldstein, Harvey, Friedman-Weieneth, et al., 2007). Preschoolers in turn may be especially sensitive to disruptions in family functioning because this is a formative period of social, emotional, and behavioral development (Denham, 2006).

These two models predict specific patterns of the early development of ADHD and ODD symptoms. The developmental precursor model suggests that early ADHD symptoms should predict later ODD symptoms, but not vice versa. The correlated risk factors model suggests that comorbidity between ADHD and ODD symptoms should emerge at an early age as a function of correlated risk factors. Family history, which is a "proxy for genetic, environmental, and behavioral risks for health" (Doerr & Teng, 2012) may be one set of key correlated risk factors. To evaluate these two models, important insights may be gained by examining the transactional relation between ADHD and ODD symptoms and evaluating the role of family history in the early development of comorbidity between symptoms.

## Transactional Relations Between Symptoms of ADHD and ODD

Longitudinal studies that evaluate whether symptoms of one disorder predict later symptoms of a second disorder can provide inroads into teasing apart the effects of one disorder on the development of another disorder. A relatively large body of research on older children has evaluated the relation between early ADHD and later conduct problems and has revealed mixed findings. A number of studies have found that ADHD symptoms in school-age children predict conduct problems in later childhood or adolescence (e.g., Diamantopoulou, Verhulst, & van der Ende, 2010; Pardini & Fite, 2010), but others have not (see Lahey, McBurnett, & Loeber, 2000, for a review). Fewer studies have examined ODD as a predictor of later ADHD, but there is some evidence supporting this component of the transactional relation (Biederman, Faraone, Milberger, Curtis, et al., 1996; Stringaris & Goodman, 2009). The few studies that have tested the bidirectional relation between symptoms of ADHD and ODD have found that ADHD symptoms predicted later ODD symptoms, but early ODD symptoms did not predict later symptoms of ADHD (Burke, Loeber, Lahey, & Rathouz, 2005; Burns & Walsh, 2002).

No studies have tested a cross-lagged model of ADHD and ODD during the preschool years. However, a handful of longitudinal studies have used regression analyses to test relevant pathways. Campbell, Breaux, Ewing, and Szumowski (1986) reported that aggression at age 3 predicted hyperactivity at age 4 (but not at age 6) controlling for age 3 hyperactivity, but hyperactivity at age 3 did not predict later aggression controlling for age 3 aggression. Lavigne and colleagues (2001) and Tandon and colleagues (2011) also found that ODD in preschool children predicted later ADHD diagnoses, but did not test whether early ADHD was predictive of later ODD. Wåhlstedt, Thorell, and Bohlin (2008) reported that ADHD symptoms in 4- to 6-year-old children predicted later ODD symptoms but did not test for or control for early ODD symptoms. Similarly, Lahey and colleagues (2009) found that hyperactivity/inattention in 4- to 7-year-old children significantly predicted later conduct problems; however, this effect was reduced when controlling for early conduct problems. Finally, Speltz, McClellan, DeKlyen, and Jones (1999) failed to find that ADHD and ODD in 4- to 5 [1/2]-year-old preschoolers raised the risk of the other disorder 2 years later. These mixed findings may be a result of differences in whether studies controlled for early symptoms, differences in the ages of children, and differences in follow-up periods. A more complete understanding of the transactional relation between ADHD and ODD symptoms can be gained by examining how ADHD and ODD predict one another over multiple time points across the preschool years.

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## Family History of ADHD and ODD

A number of studies of older children and adolescents with attention and disruptive behavior disorders have examined family histories of ADHD and conduct problems, though much of the research has focused on CD rather than ODD. Findings have consistently suggested that children with ADHD, with and without CD, have family histories of ADHD, but only children with ADHD/CD have elevated rates of family histories of CD or antisocial personality disorder (APD; Faraone, Biederman, & Monuteaux, 2000; Frick, Lahey, Christ, Loeber, & Green, 1991; Petty et al., 2009; Smalley et al., 2000). Moreover, studies have found evidence for cosegregation between family histories of ADHD and CD—that is, comorbidity between ADHD and CD in relatives of children with ADHD and CD is higher than expected by chance (Faraone et al., 2000; Petty et al., 2009). Fewer studies have examined ODD, and studies of ODD in both children and family members are particularly sparse. Whereas some studies have failed to find elevated family histories of CD or APD in children with ADHD/ODD (Faraone et al., 2000; Petty et al., 2009), others have found significant elevations (Frick et al., 1992). There is some evidence that children with ADHD/ODD have elevated rates of ODD in their families compared with nonproblem children (Faraone et al., 2000; Petty et al., 2009); however, family histories of ODD are also found in children with ADHD without ODD/CD and in children with ADHD/CD (Faraone et al., 2000).

There is a smaller body of research linking family histories of ADHD and ODD/CD with symptoms in preschoolers. Tandon et al. (2011) reported that family history of ADHD/CD was related to ADHD diagnoses in preschoolers but did not tease apart ADHD from CD/ODD. Similarly, Pauli-Pott, Dalir, Mingebach, Roller, and Becker (2013) found that family history of ADHD significantly predicted preschool ADHD symptoms, but did not take into account comorbid ODD symptoms. Galéra and colleagues (2011) found that paternal history of antisocial behavior distinguished young children who demonstrated elevated trajectories of hyperactivity and/or inattention from children with low trajectories, but did not examine family history of ADHD. In a previous study based on the present dataset, 3-year-old children with comorbid ADHD and ODD symptoms and children with only ADHD symptoms both demonstrated elevated family histories of ADHD and ODD, but only children with comorbid ADHD/ODD symptoms showed an elevated family history of CD symptoms (Harvey et al., 2007). Thus, research on older children provides substantial indirect evidence that family history of ADHD/ODD/CD may be important determinants of the early emergence of ADHD and ODD, and a small body of direct evidence with preschool children provides further corroboration.

## The Present Study

Some evidence exists in support of both the developmental precursors and the correlated risk factors models. However, the bulk of research in support of these models is based on older children and adolescents. Moreover, few studies have examined the transactional relation between ADHD and ODD over multiple time points. Finally, although researchers have suggested that multiple models may be at work (Rhee et al., 2008), the literature addressing these models has remained largely disparate. Combining these theories may provide a more comprehensive model of the development of comorbidity between ADHD and ODD. The present study seeks to advance our understanding of how comorbidity between ADHD and ODD develops by examining symptoms from age 3 to age 6 in a sample of children at risk for ADHD and ODD. A model (see Figure 1) that combines the developmental precursor model and the correlated risk factors model will be tested. In this model, family histories of ADHD and ODD/CD are posited to be correlated and to uniquely predict children's symptoms of ADHD and ODD, respectively, during the early preschool years, which in turn are predicted to be stable across the preschool years. Furthermore, ADHD at one time point is hypothesized to predict ODD symptoms at the next time point. ADHD and ODD are thought to be multidimensional (American Psychiatric Association, 2013), so this study will test models separately for inattention and hyperactivity/impulsivity and for anger/irritability and argumentative/defiance.[1]

## Method

### Participants

Participants were 199 children (107 boys, 92 girls) and their 199 female primary caregivers who took part in a longitudinal study of preschool children with behavior problems. Children were 3 years old at screening and 36 to 50 months ($M$ = 44 months, $SD$ = 3.4) at the first home visit (T1). Data used in the present study were collected from families at 1-year (T2; $M$ = 57 months), 2-year (T3; $M$ = 69 months), and 3-year (T4; $M$ = 81 months) follow-up visits. The sample included European American (49.7%), Latino/a American (21.6%; mostly Puerto Rican), African American (12.6%), and multiethnic (16.1%) children. The median family income at T1 was $47,108. Most mothers (84.4%) had high school diplomas and 33.2% had bachelor's degrees. More than half of children met criteria for ADHD or ODD at T4; of 168 children who were evaluated at T4, 36 met criteria for ADHD only (16 girls), 22 for ODD only (13 girls), and 39 for both ADHD and ODD (13 girls), supporting the notion that this was an at-risk sample (for details, see Harvey, Metcalfe, Herbert, & Fanton, 2011).

### Procedure

Children with significant externalizing problems were recruited over a 3-year period from 3-year-old children ($N$ = 1752) whose parents completed a screening packet that they received through the mail (via state birth records), pediatrician offices, child care centers, and community centers. A smaller group of typically developing children were also recruited but are not included in this study. Inclusion criteria were (a) no evidence based on parent report of intellectual disability, deafness, blindness, language delay, cerebral palsy, epilepsy, autism, or psychosis; (b) parent responded "yes" or "possibly" to the question, "Are you concerned about your child's activity level, defiance, aggression, or impulse control?" and (c) Be-

[1] The third dimension of ODD, vindictiveness, was not studied because it occurred infrequently in the sample (Harvey, Lugo-Candelas, & Breaux, 2015), and consists of only one symptom.

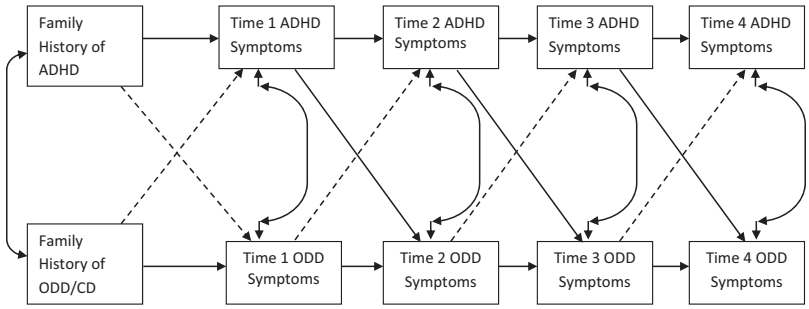


*Figure 1.* A combined correlated risk factors and developmental precursor model. Solid arrows are predicted to be significant, whereas dotted arrows are not predicted to be significant.

havior Assessment System for Children—Parent Report Scale (BASC-PRS; Reynolds & Kamphaus, 1992) Hyperactivity and/or Aggression subscale *T* scores fell at or above 65 (approximately 92nd percentile). Fifty-nine percent of behavior problem children whom we sought to recruit participated. Written informed consent was obtained from all parents who participated and parents were paid for participating. The study was conducted in compliance with the authors' Institutional Review Board.

## Measures

**Screening measure.** The BASC-PRS Preschool version assesses a broad range of psychopathology in children ages 2 [1/2] to 5 and was used at screening. *T* scores (based on general, not gender-specific, norms) for the Hyperactivity (16 items) and Aggression (13 items) subscales were used. These subscales have demonstrated good reliability and validity (Reynolds & Kamphaus, 1992). The Hyperactivity subscale consists largely of items assessing hyperactive and impulsive behavior, and also has two symptoms that more closely align with ODD (loses temper, screams). The Aggression subscale includes items assessing both verbal aggression (argumentativeness, blaming others, and teasing others) and physical aggression.

**Diagnostic interview.** At T1, T2, and T3, the ADHD and ODD sections of the NIMH-Diagnostic Interview Schedule for Children-IV (DISC-IV; Shaffer, Fisher, Lucas, Dulcan, & Schwab-Stone, 2000) were administered, with minor modification to school-related questions. The full computerized version of the DISC-IV was administered at T4. The DISC-IV is a highly structured interview in which parents respond to yes/no questions regarding presence or absence of symptoms. Interviews were administered to both mothers and fathers when they were available. Mothers' responses were used in the rare case of open disagreement between the mother and father. Symptom counts for hyperactivity/impulsivity (HI), inattention (ATT), anger/irritability (AI), and argumentative/defiance (AD) were used in the present study. Kuder–Richardson 20 ranged from .82 to .86 across time points for the nine HI symptoms, from .80 to .87 for the nine ATT symptoms, from .61 to .73 for the three AI symptoms, and from .60 to .64 for the four AD symptoms.

**Family history of behavior problems.** Parents were administered a family history interview in which they were asked whether various biological relatives of the child had problems with ADHD, ODD, and CD symptoms before the age of 18. The same caregivers who were present for the DISC interview jointly participated in this interview. The format of the scale is similar to that described by Frick et al. (1991), but was updated for *Diagnostic and Statistical Manual for Mental Disorders-Fourth Edition* (*DSM–IV*), and ODD symptoms were added. The following items were used for CD: fought with other children, vandalized, stole, skipped school, ran away from home, lied, and set fires. The parents reported the presence or absence of each symptom in both biological parents, the child's full siblings (18 months and older), and the child's biological aunts and uncles. Ratings of symptoms in aunts and uncles were used for the present study to obtain a measure of family history that was relatively independent of members of the family living in the household, whose symptoms likely influence family interactions. The number of symptoms reported for each aunt and uncle were calculated and averaged across aunts/uncles separately for HI, ATT, ODD, and CD.

## Data Analyses

Path modeling with Mplus (Muthén & Muthén, 1998–2010) was used to test the model presented in Figure 1, separately for HI, ATT, AI, and AD, and separately for family history of ODD and CD. Family histories of HI were used in HI models and family histories of ATT were used in ATT models. Family history of ADHD was separated into ATT and HI, whereas family history of ODD was not separated into AI and AD because there were a larger number of items for the two ADHD dimensions, and because there is a stronger literature base supporting the distinction between HI and ATT. Cross-lagged paths were allowed to vary across time to evaluate whether there might be developmentally sensitive periods for the development of comorbidity. The errors of ADHD and ODD symptoms were allowed to correlate between one time point and the next to account for measure-specific shared error. Correlated error terms that were significant using an $\alpha$ of .01 were retained in all subsequent models. T1 HI/ATT and AI/AD were regressed on a dummy coded race/ethnicity variable (1 = White, non-Hispanic; 0 = non-White) and family income as controls (these paths are not included in figures displaying results for simplicity). Model fit was evaluated by using four indicators: $\chi^2/df$ ($<2$ indicates good model fit), root mean square error of approximation (RMSEA; values of .08 and lower represent acceptable model fit and values between .08 and .1 indicate mediocre model

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fit), Bentler's Comparative Fit Index (CFI; values higher than .90 indicate acceptable model fit), and standardized root mean square residual (SRMR; values lower than .08 indicate adequate model fit).

Diagnostic interview data were missing for .5% of children at T1, 9% of children at T2, 30% of children at T3, and 18.5% of children at T4. Family history data were missing from 5.5% of children. Analyses comparing children who were missing data at T2, T3, and T4 to children who had data at those time points indicated no significant differences on T1 HI, ATT, AD, or AI (all $ps > .09$). There were also no significant differences on family income (all $ps > .18$), maternal education (all $ps > .06$), or gender (all $ps > .05$). There were significant racial/ethnic differences between children who were and were not missing diagnostic interview data at T4, $\chi^2(1) = 9.39$, $p = .002$; children who were missing data were less likely to be European American (27%) than children who were not missing data (55%). There were not differences on race/ethnicity at T2 or T3 ($ps > .33$). These attrition analyses suggest that missing data are most consistent with being missing at random. Full information maximum likelihood was used to address missing data. In this method, all observed information (including from cases with some missing data) is used to estimate parameters. This method has been shown to yield unbiased estimates when data are missing at random (Enders & Bandalos, 2001).

## Results

Table 1 presents descriptive statistics and intercorrelations for all study variables.

### Model Construction

**HI and AI models.** First, a set of parallel trait autoregressive models were fit to the data. HI and AI scores at each time point were regressed on their respective scores one year earlier. The concurrent correlations between HI and AI and between family history variables were also included in the model. These models demonstrated generally adequate fit both using family history of ODD, $\chi^2/df = 1.80$, RMSEA = .07, CFI = .92, SRMR = .10, and family history of CD, $\chi^2/df = 2.00$, RMSEA = .07, CFI = .92, SRMR = .10. Next, paths were added from family history of ODD (or CD) to T1 AI and from family history of HI to T1 HI. Adding these paths significantly improved model fit for the family history of ODD model, $\Delta\chi^2(2) = 12.32$, $p = .002$, and family history of CD model, $\Delta\chi^2(2) = 12.51$, $p = .002$, so they were retained in subsequent models. Adding paths from family history of ODD (CD) to T1 HI and from family history of HI to T1 AI did not significantly improve fit, $\Delta\chi^2(2) = 0.76$, $p = .68$. Next, to test whether stability of HI and AI varied across ages, autoregressive paths were constrained to be similar across time for each dimension. Fixing autoregressive paths did not result in significantly worse model fit for AI, $\Delta\chi^2(2) = 1.28$, $p = .53$, but resulted in marginally significantly worse model fit for HI, $\Delta\chi^2(2) = 5.26$, $p = .08$. Therefore, autoregressive paths were fixed across time points for AI, but allowed to vary across time for HI.

Cross-lagged paths from HI to AI were then added. Constraining the cross-lagged paths to be equal across time resulted in marginally significantly worse fit, $\Delta\chi^2(2) = 5.39$, $p = .07$, so cross-lagged paths from HI to AI were allowed to vary across time. The addition of cross-lagged paths from HI to AI (free across time) marginally significantly improved model fit compared with the parallel trait model, $\Delta\chi^2(3) = 6.98$, $p = .07$. Inspection of path coefficients indicated that the path from HI to AI was significant only from T3 to T4. Therefore, cross paths from HI to AI from T1 to T2 and from T2 to T3 were dropped. The resulting model significantly improved fit compared with the parallel trait model, $\Delta\chi^2(1) = 4.97$, $p = .03$. Next, cross-lagged paths from AI to HI were added to the parallel trait model. Constraining the cross-lagged paths to be equal across time did not result in significantly worse fit, $\Delta\chi^2(2) = 3.54$, $p = .17$. The addition of cross-lagged paths from AI to HI (fixed across time) did not significantly improve model fit compared with the parallel trait model, $\Delta\chi^2(1) = 0.14$, $p = .71$. Therefore, the model with paths from T3 HI to T4 AI was selected as the final model (see Figure 2). This model demonstrated adequate fit both using family history of ODD, $\chi^2/df = 1.50$, RMSEA = .05, CFI = .94, SRMR = .09, and family history of CD, $\chi^2/df = 1.71$, RMSEA = .06, CFI = .92, SRMR = .10.

**HI and AD models.** The parallel trait autoregressive models for HI and AD demonstrated generally adequate fit both using family history of ODD, $\chi^2/df = 1.96$, RMSEA = .07, CFI = .91, SRMR = .11, and family history of CD, $\chi^2/df = 2.10$, RMSEA = .07, CFI = .89, SRMR = .11. Adding paths from family history of ODD (or CD) to T1 AD and from family history of HI to T1 HI significantly improved model fit for the family history of ODD model, $\Delta\chi^2(2) = 8.12$, $p = .02$, and family history of CD model, $\Delta\chi^2(2) = 8.02$, $p = .02$. Adding paths from family history of ODD (CD) to T1 HI and from family history of HI to T1 AD did not significantly improve fit for the family history of CD model, $\Delta\chi^2(2) = 0.81$, $p = .67$, or for the family history of ODD models, $\Delta\chi^2(2) = 0.70$, $p = .70$. Thus, paths from family history of ODD (CD) to T1 AD and from family history of HI to T1 HI were retained in subsequent models. Fixing autoregressive paths did not result in significantly worse model fit for AD, $\Delta\chi^2(2) = .14$, $p = .93$, but resulted in marginally significantly worse model fit for HI, $\Delta\chi^2(2) = 5.27$, $p = .07$. Therefore, autoregressive paths for AD were fixed across time points and autoregressive paths for HI were allowed to vary across time.

Constraining cross-lagged paths from HI to AD to be equal across time did not result in significantly worse fit, $\Delta\chi^2(2) = 1.87$, $p = .39$. The addition of these paths (fixed across time) significantly improved model fit compared with the parallel trait model, $\Delta\chi^2(1) = 9.37$, $p = .002$. Constraining the cross-lagged paths from AD to HI to be equal across time also did not result in significantly worse fit, $\Delta\chi^2(2) = 2.37$, $p = .31$, and adding these paths (fixed across time) did not significantly improve model fit compared with the parallel trait model, $\Delta\chi^2(1) = 1.62$, $p = .20$. Therefore, the model with paths from HI to AD (fixed across time) was selected as the final model (see Figure 3). This model demonstrated adequate fit both using family history of ODD, $\chi^2/df = 1.64$, RMSEA = .06, CFI = .93, SRMR = .09, and family history of CD, $\chi^2/df = 1.79$, RMSEA = .06, CFI = .92, SRMR = .09.

**ATT and AI models.** Parallel trait autoregressive models for ATT and AI demonstrated generally adequate fit both using family history of ODD, $\chi^2/df = 1.42$, RMSEA = .06, CFI = .96, SRMR = .10, and family history of CD, $\chi^2/df = 1.65$, RMSEA = .06, CFI = .93, SRMR = .10. Adding paths from family history of

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Table 1
*Correlation Among Family History, ADHD and ODD Symptoms, and Control Variables*

| Variable | Mean (*SD*) | *N* | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Family history HI | 1.65 (1.45) | 188 | — | | | | | | | | | | | | | | | | | | | | | |
| 2. Family history ATT | 1.76 (1.57) | 188 | .70$^{***}$ | — | | | | | | | | | | | | | | | | | | | | |
| 3. Family history ODD | 2.00 (1.60) | 188 | .57$^{***}$ | .64$^{***}$ | — | | | | | | | | | | | | | | | | | | | |
| 4. Family history CD | 0.60 (0.76) | 188 | .43$^{***}$ | .40$^{***}$ | .57$^{***}$ | — | | | | | | | | | | | | | | | | | | |
| 5. T1 HI | 5.48 (2.08) | 198 | .23$^{**}$ | .14$^{\dagger}$ | .18$^{*}$ | .21$^{**}$ | — | | | | | | | | | | | | | | | | | |
| 6. T1 ATT | 4.20 (2.58) | 198 | .16$^{*}$ | .21$^{**}$ | .24$^{**}$ | .22$^{**}$ | .60$^{***}$ | — | | | | | | | | | | | | | | | | |
| 7. T1 AI | 1.86 (1.04) | 198 | .10 | .13$^{\dagger}$ | .19$^{*}$ | .22$^{**}$ | .28$^{***}$ | .33$^{***}$ | — | | | | | | | | | | | | | | | |
| 8. T1 AD | 2.36 (0.97) | 198 | .05 | .07 | .05 | .07 | .32$^{***}$ | .38$^{***}$ | .54$^{***}$ | — | | | | | | | | | | | | | | |
| 9. T2 HI | 5.34 (2.23) | 181 | .18$^{*}$ | .19$^{*}$ | .18$^{*}$ | .15$^{\dagger}$ | .54$^{***}$ | .39$^{***}$ | .14$^{\dagger}$ | .19$^{*}$ | — | | | | | | | | | | | | | |
| 10. T2 ATT | 3.93 (2.67) | 181 | .17$^{*}$ | .18$^{*}$ | .19$^{*}$ | .15$^{*}$ | .40$^{***}$ | .56$^{***}$ | .17$^{**}$ | .26$^{***}$ | .57$^{***}$ | — | | | | | | | | | | | | |
| 11. T2 AI | 1.75 (1.11) | 182 | .11 | .14$^{\dagger}$ | .10 | .12 | .17$^{*}$ | .18$^{*}$ | .43$^{***}$ | .25$^{**}$ | .30$^{***}$ | .26$^{**}$ | — | | | | | | | | | | | |
| 12. T2 AD | 2.25 (1.12) | 182 | .19$^{*}$ | .19$^{*}$ | .16$^{*}$ | .15$^{\dagger}$ | .16$^{*}$ | .21$^{**}$ | .28$^{***}$ | .39$^{***}$ | .37$^{***}$ | .38$^{***}$ | .56$^{***}$ | — | | | | | | | | | | |
| 13. T3 HI | 4.97 (2.65) | 140 | .25$^{**}$ | .26$^{**}$ | .29$^{**}$ | .33$^{***}$ | .43$^{***}$ | .32$^{***}$ | .17$^{*}$ | .11 | .65$^{***}$ | .54$^{***}$ | .25$^{**}$ | .39$^{***}$ | — | | | | | | | | | |
| 14. T3 ATT | 3.95 (2.90) | 140 | .18$^{*}$ | .21$^{*}$ | .14 | .11 | .20$^{*}$ | .42$^{***}$ | .18$^{*}$ | .20$^{*}$ | .48$^{***}$ | .66$^{***}$ | .27$^{**}$ | .39$^{***}$ | .61$^{***}$ | — | | | | | | | | |
| 15. T3 AI | 1.52 (1.03) | 140 | .12 | .19$^{*}$ | .16$^{\dagger}$ | .16$^{\dagger}$ | .18$^{*}$ | .11 | .22$^{*}$ | .09 | .10 | .20$^{*}$ | .46$^{***}$ | .30$^{***}$ | .31$^{***}$ | .27$^{**}$ | — | | | | | | | |
| 16. T3 AD | 2.04 (1.21) | 140 | .08 | .19$^{*}$ | .10 | .12 | .18$^{*}$ | .13 | .18$^{*}$ | .20$^{*}$ | .28$^{**}$ | .29$^{**}$ | .41$^{***}$ | .46$^{***}$ | .50$^{***}$ | .50$^{***}$ | .56$^{***}$ | — | | | | | | |
| 17. T4 HI | 3.77 (2.61) | 162 | .18$^{*}$ | .17$^{*}$ | .16$^{*}$ | .15$^{\dagger}$ | .43$^{***}$ | .31$^{***}$ | .06 | .09 | .57$^{***}$ | .50$^{***}$ | .12 | .23$^{**}$ | .64$^{***}$ | .57$^{***}$ | .16 | .38$^{***}$ | — | | | | | |
| 18. T4 ATT | 2.89 (2.87) | 162 | .23$^{**}$ | .26$^{**}$ | .13 | .05 | .23$^{**}$ | .34$^{***}$ | .05 | .14$^{\dagger}$ | .43$^{***}$ | .49$^{***}$ | .18$^{*}$ | .30$^{***}$ | .40$^{***}$ | .67$^{***}$ | .14 | .31$^{***}$ | .60$^{***}$ | — | | | | |
| 19. T4 AI | 1.10 (1.16) | 162 | .16 | .22$^{**}$ | .27$^{**}$ | .29$^{***}$ | .06 | .15$^{\dagger}$ | .24$^{**}$ | .12 | .21$^{**}$ | .21$^{**}$ | .37$^{***}$ | .29$^{***}$ | .30$^{***}$ | .21$^{*}$ | .46$^{***}$ | .32$^{***}$ | .27$^{**}$ | .26$^{**}$ | — | | | |
| 20. T4 AD | 1.36 (1.19) | 162 | .17$^{*}$ | .22$^{**}$ | .26$^{**}$ | .14$^{\dagger}$ | .16$^{*}$ | .17$^{*}$ | .15$^{\dagger}$ | .17$^{*}$ | .27$^{**}$ | .26$^{**}$ | .27$^{***}$ | .34$^{***}$ | .43$^{***}$ | .34$^{***}$ | .32$^{***}$ | .45$^{***}$ | .47$^{***}$ | .38$^{***}$ | .66$^{***}$ | — | | |
| 21. Gender[a] | .46 | 199 | .04 | .03 | .13$^{\dagger}$ | .07 | −.10 | .03 | −.04 | −.01 | −.10 | .07 | .04 | .04 | −.08 | −.04 | .00 | −.10 | −.06 | −.08 | .01 | .04 | — | |
| 22. Race/ethnicity[b] | .50 | 199 | −.05 | .02 | .03 | −.22$^{**}$ | −.23$^{**}$ | −.28$^{***}$ | −.13 | −.12 | −.29$^{***}$ | −.30$^{***}$ | −.09 | −.13$^{\dagger}$ | −.32$^{***}$ | −.24$^{**}$ | −.07 | −.12 | −.18$^{*}$ | −.14$^{\dagger}$ | −.05 | −.04 | −.14$^{\dagger}$ | — |
| 23. Income (in $1,000s) | 54.42 (38.51) | 199 | −.15$^{*}$ | −.11 | −.15$^{*}$ | −.20$^{**}$ | −.24$^{**}$ | −.28$^{***}$ | −.14 | −.08 | −.29$^{***}$ | −.28$^{***}$ | −.08 | −.07 | −.36$^{***}$ | −.26$^{**}$ | −.16$^{\dagger}$ | −.19$^{*}$ | −.22$^{**}$ | −.16$^{*}$ | −.12 | −.09 | −.08 | .26$^{***}$ |

*Note*. ADHD = attention-deficit/hyperactivity disorder; HI = hyperactivity/impulsivity; ATT = inattention; ODD = oppositional defiant disorder; CD = conduct disorder; AI = anger/irritability; AD = anger/defiance.
[a] Gender was coded 1 for female and 0 for male. [b] Race/ethnicity was coded 1 for white/European American and 0 for all of the races/ethnicities.
$^{\dagger}$ $p < .10$. $^{*}$ $p < .05$. $^{**}$ $p < .01$. $^{***}$ $p < .001$.

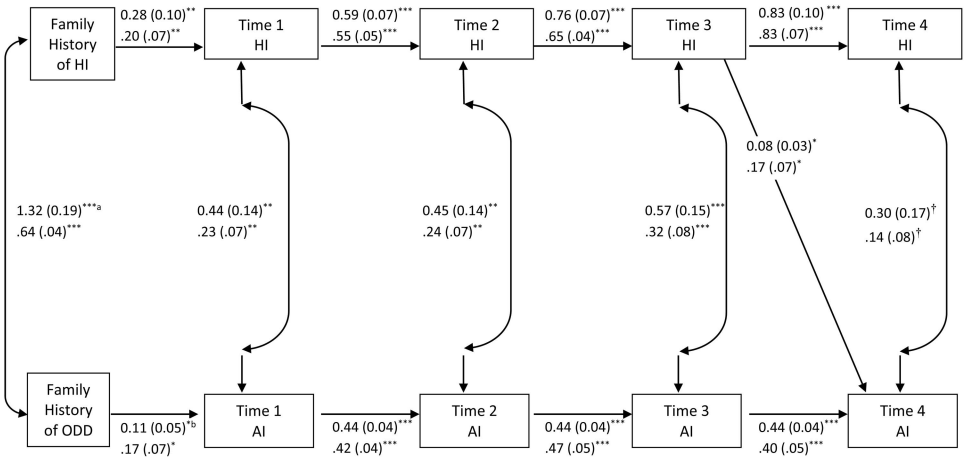


*Figure 2.* A combined correlated risk factors and developmental precursor model for symptoms of hyperactivity/impulsivity and anger/irritability. Unstandardized coefficients are on top and standardized coefficients are on the bottom. ODD = oppositional defiant disorder; CD = conduct disorder; HI = hyperactivity/impulsivity; AI = anger/irritability. [a] For family history of CD the unstandardized coefficient was 1.32 (0.19), $p < .001$, and standardized was .64 (.04), $p < .001$. [b] For family history of CD the unstandardized coefficient was 0.24 (0.10), $p < .05$, and standardized was .18 (.07), $p < .05$. $^{*}$ $p < .05$, $^{**}$ $p < .01$, $^{***}$ $p < .001$, $^{\dagger}$ $p < .10$.

ODD (or CD) to AD scores at T1 and from family history of ATT to T1 ATT significantly improved model fit for both family history of ODD models, $\Delta\chi^2(2) = 9.78$, $p = .01$, and for family history of CD models, $\Delta\chi^2(2) = 10.82$, $p = .004$. Adding paths from family history of ODD (CD) to T1 ATT and from family history of ATT to T1 AI did not significantly improve fit for the family history of CD models, $\Delta\chi^2(2) = 1.50$, $p = .47$, or for the family history of ODD models, $\Delta\chi^2(2) = 3.39$, $p = .18$. Thus, paths from family history of ODD (CD) to T1 AI and paths from family history of ATT to T1 ATT were retained in subsequent models. Fixing autoregressive paths for AI did not result in significantly worse model fit, $\Delta\chi^2(2) = 0.91$, $p = .63$, nor did fixing autoregressive paths for ATT, $\Delta\chi^2(2) = 1.13$, $p = .57$, so autoregressive paths for AI and ATT were fixed across time points.

Constraining cross-lagged paths from ATT to AI to be equal across time did not result in significantly worse fit, $\Delta\chi^2(2) = 1.32$, $p = .52$. The addition of these paths (fixed across time) marginally improved model fit compared with the parallel trait model, $\Delta\chi^2(1) = 3.14$, $p = .08$. Constraining cross-lagged paths from AD to ATT to be equal across time did not result in significantly worse

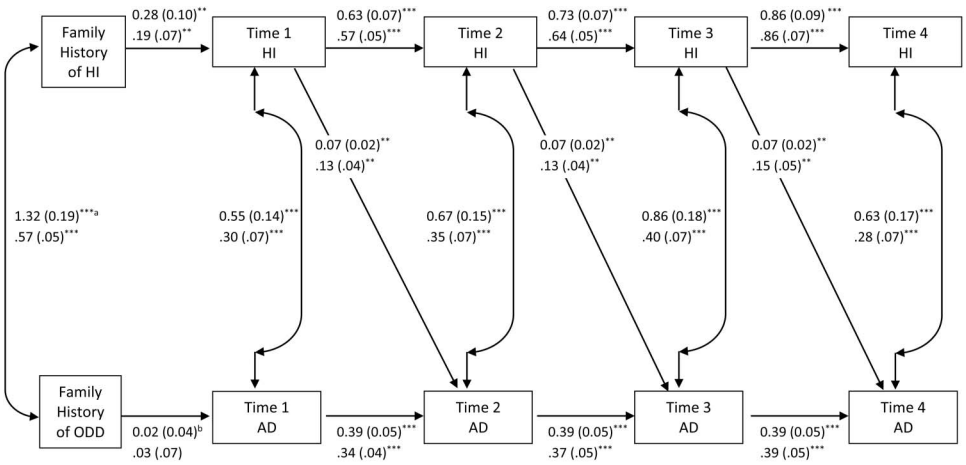


*Figure 3.* A combined correlated risk factors and developmental precursor model for symptoms of hyperactivity/impulsivity and argumentative/defiance. Unstandardized coefficients are on top and standardized coefficients are on the bottom. ODD = oppositional defiant disorder; CD = conduct disorder; HI = hyperactivity/impulsivity; AD = argumentative/defiant. [a] For family history of CD the unstandardized coefficient was 0.47 (0.09), $p < .001$, and standardized coefficient was .43 (.06), $p < .001$. [b] For family history of CD the unstandardized coefficient was 0.02 (0.09), $p > .05$, and standardized coefficient was .02 (.07), $p > .05$. $^{*}$ $p < .05$, $^{**}$ $p < .01$, $^{***}$ $p < .001$.

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fit, $\Delta\chi^2(2) = 1.06$, $p = .59$. The addition of these paths (fixed across time) did not significantly improve model fit compared with the parallel trait model, $\Delta\chi^2(1) = .42$, $p = .52$. Thus, the model with paths from ATT to AI fixed across time was chosen as the final model (see Figure 4). This model demonstrated adequate fit for both family history of ODD and CD models, $\chi^2/df = 1.18$, RMSEA = .03, CFI = .98, SRMR = .07.

**ATT and AD models.** Parallel trait autoregressive models for ATT and AD demonstrated generally adequate fit both using family history of ODD, $\chi^2/df = 1.54$, RMSEA = .05, CFI = .95, SRMR = .11, and family history of CD, $\chi^2/df = 1.63$, RMSEA = .06, CFI = .93, SRMR = .11. Adding paths from family history of ODD (or CD) to AD scores at T1 and from family history of ATT to T1 ATT significantly improved model fit for both family history of ODD models, $\Delta\chi^2(2) = 6.58$, $p = .04$, and for family history of CD models, $\Delta\chi^2(2) = 6.61$, $p = 04$. Adding paths from family history of ODD (CD) to T1 ATT and from family history of ATT to T1 AD did not significantly improve fit for the family history of CD models, $\Delta\chi^2(2) = 1.67$, $p = .43$, or for the family history of ODD models, $\Delta\chi^2(2) = 4.55$, $p = .11$. Thus, paths from family history of CD/ODD to T1 AD and paths from family history of ATT to T1 ATT were retained in subsequent models. Fixing autoregressive paths did not result in significantly worse model fit for AD, $\Delta\chi^2(2) = .05$, $p = .98$, or for ATT, $\Delta\chi^2(2) = .80$, $p = .67$, so these paths were fixed across time points.

Constraining the cross-lagged paths from ATT to AD to be equal across time did not result in significantly worse fit, $\Delta\chi^2(2) = .52$, $p = .77$. The addition of these paths (fixed across time) significantly improved model fit compared with the parallel trait model, $\Delta\chi^2(1) = 8.36$, $p = .003$. Constraining the cross-lagged paths from AD to ATT to be equal across time did not result in significantly worse fit, $\Delta\chi^2(2) = 1.34$, $p = .51$. The addition of these paths (fixed across time) significantly improved model fit compared with the parallel trait model, $\Delta\chi^2(1) = 4.60$, $p = .03$. The model with paths from ATT to AD and from AD to ATT fixed across time fit the data significantly better than the parallel trait model, $\chi^2(2) = 11.40$, $p = .003$, and was selected as the final model (see Figure 5). This model demonstrated adequate fit both using family history of ODD, $\chi^2/df = 1.20$, RMSEA = .03, CFI = .98, SRMR = .08, and family history of CD, $\chi^2/df = 1.29$, RMSEA = .04, CFI = .97, SRMR = .08.

## Examination of Sex as a Moderator of Family History and Cross-Lagged Paths

To examine whether models were different for boys and girls, multigroup analyses were used. When each of the cross-lagged paths and each of the paths from family history of HI/ATT and ODD/CD to T1 HI/ATT and AI/AD were constrained to be equal for boys and girls, model fit was significantly different for the ATT/AD models for family history of ODD, $\Delta\chi^2$ (10) = 22.46, $p = .01$, and approached significance for the ATT/AD models for family history of CD, $\Delta\chi^2$ (10) = 17.58, $p = .06$, suggesting that parameters were significantly different for boys than for girls. Follow-up models separately fixing paths from ATT to AD, from AD to ATT, and family history paths indicated that the gender differences were driven by differences in paths from AD to ATT, $\Delta\chi^2(3) = 9.05$, $p = .02$. Paths from AD at one time point to ATT at the next time point were significant for boys, $b = .33$, $SE = .16$, $p = .04$, but not for girls, $b = .07$, $SE = .14$, $p = .64$.

## Summary of Support for the Developmental Precursors and Correlated Risk Factors Models

The final models provided support for the developmental precursors model, particularly for AD. There was no evidence that ODD symptoms significantly predicted HI symptoms at the next time point, controlling for current HI symptoms. ODD symptoms were also not significantly predictive of ATT symptoms, though paths from AD to later ATT approached significance, and were

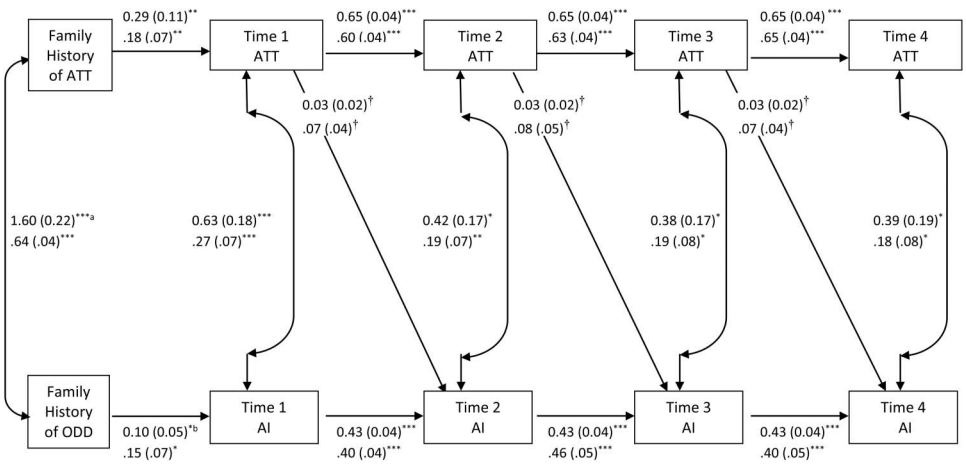


*Figure 4.* A combined correlated risk factors and developmental precursor model for symptoms of inattention and anger/irritability. Unstandardized coefficients are on top and standardized coefficients are on the bottom. ODD = oppositional defiant disorder; CD = conduct disorder; ATT = inattention; AI = anger/irritability. [a] For family history of CD the unstandardized coefficient was 0.48 (0.09), $p < .001$, and standardized coefficient was .40 (.06), $p < .001$. [b] For family history of CD the unstandardized coefficient was 0.23 (0.10), $p < .05$, and standardized was .17 (.07), $p < .05$. * $p < .05$, ** $p < .01$, *** $p < .001$, † $p < .10$.

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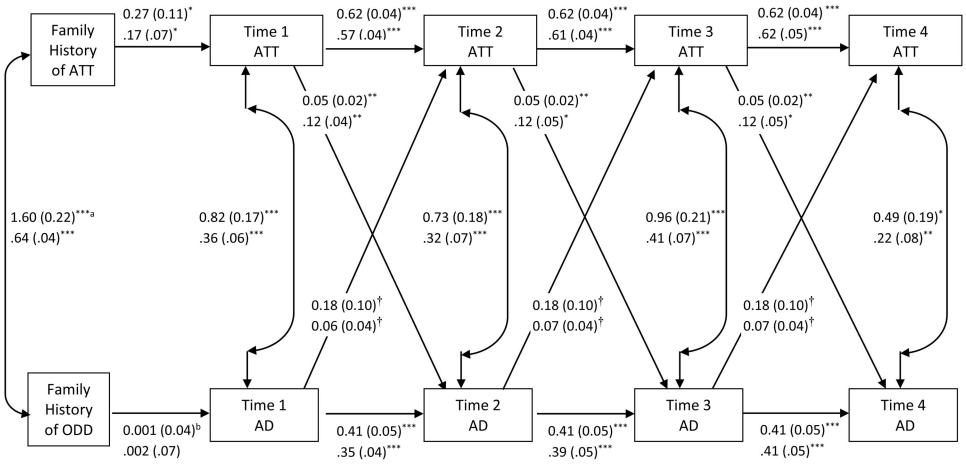


*Figure 5.* A combined correlated risk factors and developmental precursor model for symptoms of inattention and argumentative/defiance. Unstandardized coefficients are on top and standardized coefficients are on the bottom. ODD = oppositional defiant disorder; CD = conduct disorder; ATT = inattention; AD = argumentative/defiant. [a] For family history of CD the unstandardized coefficient was 0.48 (0.09), $p < .05$, and standardized was .40 (.06), $p < .05$. [b] For family history of CD the unstandardized coefficient was 0.02 (0.09), $p > .05$, and standardized was .01 (.07), $p > .05$. * $p < .05$, ** $p < .01$, *** $p < .001$, † $p < .10$.

significant for boys. In contrast, at each time point, HI significantly predicted AI at the next time point, and T3 HI significantly predicted T4 AD. Similarly, at each time point, paths from ATT to AD/AI at the next time point were significant for AD and approached significance for AI. The final models also provided support for the correlated risk factors model for AI, but not for AD. Family history of HI and ATT were both significantly correlated with family histories of ODD and CD. Family history of HI/ATT significantly predicted T1 HI/ATT, but did not significantly predict T1 AI/AD. Family histories of ODD and CD both significantly predicted T1 AI, but did not significantly predict T1 AD, HI, or ATT.

## Discussion

The present study sought to test a model that integrated two previously proposed models of the development of comorbidity between ADHD and ODD symptoms. In keeping with the developmental precursors model, we examined whether early symptoms of ADHD predict later ODD symptoms, but not vice versa. In addition, we examined whether family history of ADHD and ODD/CD represent correlated risk factors that predict early symptoms of ADHD and ODD in children, thus accounting in part for their comorbidity. Results provided support for both components of the model, suggesting that multiple mechanisms are likely involved, though results varied across dimensions of ODD.

### Developmental Precursors Model

The developmental precursors model was most clearly supported for explaining the association between ADHD symptoms and argumentative/defiant symptoms. This model posits that symptoms of ADHD disrupt family functioning, which in turn contributes to the development of ODD symptoms. As predicted, symptoms of ADHD predicted future argumentative/defiant symptoms, controlling for concurrent ADHD. There was only weak evidence that the developmental precursors model plays a role in the development of comorbidity between ADHD and anger/irritability; paths from ADHD symptoms to future anger/irritability approached but did not reach significance for inattention and were only significant for hyperactivity/impulsivity at the final time point.

ODD symptoms did not generally predict future ADHD symptoms, controlling for concurrent ADHD; however, there was evidence that argumentative/defiance was predictive of future inattention for boys but not for girls. This finding is consistent with Stringaris and Goodman (2009) who found that headstrong symptoms predicted later ADHD diagnoses in older children, and theorized that delay aversion may underlie both argumentative/defiant symptoms and ADHD symptoms. However, it is unclear why this effect would be observed for boys but not for girls. Girls with ADHD tend to have less severe ADHD symptomatology than boys (Gershon, 2002), but correlates of ADHD are generally similar for boys and girls (Bauermeister et al., 2007). It may be that argumentative/defiant symptoms affect parental perceptions of inattention differently for boys and for girls. For example, previous research found that when teachers rated confederate boys who were trained to exhibit ODD symptoms, they gave higher ratings of inattention than when they rated confederate girls who were exhibiting ODD symptoms (Jackson & King, 2004). The same negative halo effect for boys may also be present in parent ratings of behavior, resulting in argumentative/defiance predicting future ratings of inattention.

Studies of older children that have tested transactional models between ADHD and ODD (Burke et al., 2005; Burns & Walsh, 2002) suggest that the effect of ADHD on ODD symptoms observed in the present study likely continues as children continue through elementary school. Results of this study stand in contrast to previous studies of preschoolers that have found that hyperac-

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tivity in 3- and 4-year-old children did not predict later aggression (Campbell et al., 1986) or ODD (Speltz et al., 1999). Campbell and colleagues (1986) used a measure of aggression that included items of physical aggression, which may account for their differing findings. Our study together with Campbell and colleagues' research suggest that ADHD symptoms may be developmental precursors to some conduct problem symptoms more than others.

Research is needed to identify the causal mechanisms through which ADHD leads to the development of ODD. Studies suggest that family and peer relationships may be important mediators of this relation. Preschoolers with ADHD symptoms are more likely than their typically developing counterparts to be exposed to negative parenting practices, parent psychopathology, family stress, and peer rejection (Johnston & Mash, 2001; Keown & Woodward, 2006). However, this literature is largely cross-sectional, making it difficult to tease apart the causal directions involved in these relations. Longitudinal examinations of such mediational pathways are needed.

## Correlated Risk Factors Model

Results provided clear support for the notion that family histories of ADHD and ODD/CD symptoms represent correlated risk factors for ADHD and anger/irritability symptoms in young children. This model suggests that early emergence of comorbidity between ADHD and ODD may be due at least in part to the fact that children with family histories of ADHD also tend to have family histories of ODD/CD and that each of these risk factors independently contributes to ADHD and anger/irritability, respectively. Our finding that there were no cross-lagged relations involved for family histories of ADHD and ODD/CD (i.e., family history of ADHD did not predict child ODD symptoms and family history of ODD/CD did not predict child ADHD symptoms) suggests that it is not the case that ADHD and ODD symptoms in children are correlated because they share family history risk factors. Instead, these symptoms each have their own unique family history predictors, which are correlated with one another. It may be that these risk factors are correlated because of a shared set of biological processes. This notion is supported by twin studies suggesting that comorbidity between ADHD and ODD/CD is largely because of shared genetics (Eaves et al., 2000; Silberg et al., 1996). However, research has not been able to determine the exact mechanism underlying shared genetics. There is clear evidence that both ADHD and ODD/CD are polygenic disorders, and that there is overlap in the genes involved in these disorders (e.g., Comings et al., 2000). Thus, the observed correlations between family histories of ADHD and ODD/CD in the present study may be a result of overlap in sets of genes involved in these two disorders. Furthermore, there may be interactions among genes, such that phenotypic expression of one gene depends on the presence of other genes, resulting in observed correlations between the two phenotypes. For example, the catechol-O-methyltransferase gene (COMT) has been strongly linked to antisocial behavior among children with ADHD, but not among children without ADHD (Caspi et al., 2008), suggesting that COMT results in antisocial behavior only in the presence of genes linked with ADHD. Research is needed to explore these possible mechanisms.

Our finding that family history of CD predicted one dimension of children's ODD symptoms (anger/irritability) is consistent with previous research suggesting that ODD and CD are caused by the same underlying liability (Eaves et al., 2000), though other studies have suggested otherwise (Petty et al., 2009). The fact that previous studies have yielded conflicting findings regarding the link between ODD and family history of CD or APD (Biederman, Faraone, Milberger, Jetton, et al., 1996; Faraone et al., 2000; Frick et al., 1992) could be because ODD symptom expression may have different causes at different developmental stages. For example, early expression of ODD may be more likely caused by genetic factors (and therefore, linked with family history of CD), whereas expression of ODD in later childhood or adolescence may be more influenced by environmental causes. In fact, studies that have found a link between ODD in children and family histories of CD/antisocial behavior (e.g., Frick et al., 1992) have focused on younger children than have studies that have failed to find a link (Biederman, Faraone, Milberger, Jetton, et al., 1996; Faraone et al., 2000). In addition, given our findings that family history of CD was linked only to anger/irritability and not argumentative/defiance, conflicting findings across studies could be because of differences in symptom expression, with children in some studies characterized more by argumentative/defiance and others more by anger/irritability. Further research is needed to examine developmental and dimensional differences in etiological correlates of ODD in children.

## Multiple Mechanisms

Our finding that both the correlated risk factors model and developmental precursors model were supported lends further evidence to the notion that multiple mechanisms are involved in the development of comorbidity between ADHD and ODD/CD. Moreover, results suggest that the correlated risk factors model may best explain the development of comorbidity between ADHD symptoms and anger/irritability, whereas the developmental precursors model may better explain the development of comorbidity between ADHD symptoms and argumentative/defiance. Few studies that address these two models have disentangled dimensions of ODD. A small body of research has examined differential heritability of different dimensions of ODD—of relevance to the correlated risk factors model—and findings have been mixed. For example, Yeh and colleagues (Yeh, Coccaro, & Jacobson, 2010) found that heritability was higher for temper tantrums than for verbal aggression. In contrast, Kuny and colleagues (2013) did not generally find differences in heritability between children exhibiting irritability versus defiance, though heritability estimates were somewhat higher for defiance among younger children. It is also important to note that the developmental precursors model and correlated risk factors model likely interact with one another. For example, the serotonin transporter gene has been found to moderate the effects of family stress on conduct problems, and this has been demonstrated in preschool-aged children (Lavigne et al., 2013) and among children with ADHD (Sonuga-Barke et al., 2009). More research is needed to better understand the additive, synergistic, and interactive effects of different mechanisms underlying the development of comorbidity between ADHD and ODD/CD.

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## Developmental Considerations

High activity level, inattention, impulsivity, anger, and defiance are common in typically developing preschoolers (Campbell, 2002). Approximately half of the children who exhibited behavior problems at age 3 in the present sample later met criteria for ADHD or ODD. Thus, ADHD and ODD symptoms in this sample represented clinical disorders for some children and transient preschool behavior problems for others. Children who later met criteria for ADHD or ODD in this sample had more severe symptomatology at age 3, 4, and 5 (Harvey, Youngwirth, Thakar, & Errazuriz, 2009). That is, in the present sample, symptom severity was a marker for clinically significant/stable problems versus transient ones. Therefore, the relations observed in this study between ADHD and ODD symptoms reflect the fact that preschool children with more severe symptoms of one condition (that likely represented an underlying clinical condition) were more likely to have severe symptoms of the other condition, compared with children with subclinical or mild symptoms (that likely represented transient preschool problems). Thus, the at-risk sample used in the present study provides insights into how comorbidity between clinically significant levels of ADHD and ODD arise relative to transient, subclinical difficulties.

## Limitations

These findings should be interpreted in the context of several limitations of this study. First, symptom ratings were based solely on diagnostic interviews with parents. Although symptom ratings based on diagnostic interviews are typically central to assessment of ADHD and ODD, future research using multiple methods of assessment will be important. Second, family history of behavior problems was assessed using parents' retrospective reports of family members. Because parents reported both on their children's symptoms and family members' symptoms, observed relations may be inflated by method variance. Similarly, parents' experiences with their own children may have colored their retrospective reports of childhood symptoms of aunts and uncles. Third, a highly structured diagnostic interview was used to assess ADHD and ODD symptoms. Because this interview does not involve clinician judgments, it is highly susceptible to method variance associated with parent report. Fourth, because this study focused only on children who were exhibiting externalizing problems at age 3, results of this study may not generalize to children with later onset of ADHD or ODD symptoms. Similarly, because of the recruitment method and later onset of ADHD predominantly inattentive presentation, few children in the study exhibited this subtype, further limiting generalization. Fifth, the study did not address the developmental mechanisms underlying these models. Future research is needed to better understand the specific processes involved in the correlated risk factors and developmental precursors models. Finally, although the sample used in the present study was ethnically diverse, there were not sufficient numbers of each ethnic group to determine whether ethnicity might moderate the observed relations.

## Implications and Future Directions

Despite these limitations, the present study offers insight into the early development of comorbidity between ADHD and ODD. Our findings support the notion that ADHD may be causally related to ODD, but not vice versa. Further research is needed to determine whether early intervention with young preschool children who are at risk for ADHD may help prevent the development of comorbid ODD. A number of parent training programs have shown to be effective in decreasing symptoms of ADHD and ODD with preschoolers with ADHD (Bor, Sanders, & Markie-Dadds, 2002; Herbert, Harvey, Roberts, Wichowski, & Lugo-Candelas, 2013; Pisterman et al., 1992), supporting the potential benefit of early intervention. It is also clear, however, that comorbidity has already begun to develop as early as age 3. Results of this study suggest that it may be possible to identify children who are at risk for developing comorbid ADHD and ODD at a very young age based on family history patterns. Because children with comorbid ADHD and ODD have substantially poorer prognoses than children with either disorder alone (Waschbusch, 2002), intervening at a very early age may have substantial benefits.

## References

American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders, fifth edition*. Arlington, VA: American Psychiatric Association.

Barkley, R. A. (2006). *Attention-deficit hyperactivity disorder: A handbook for diagnosis and treatment* (3rd ed.). New York, NY: Guilford Press.

Bauermeister, J. J., Shrout, P. E., Chávez, L., Rubio-Stipec, M., Ramírez, R., Padilla, L., . . . Canino, G. (2007). ADHD and gender: Are risks and sequela of ADHD the same for boys and girls? *Journal of Child Psychology and Psychiatry, 48,* 831–839. http://dx.doi.org/10.1111/j.1469-7610.2007.01750.x

Beauchaine, T. P., Hinshaw, S. P., & Pang, K. L. (2010). Comorbidity of attention-deficit/hyperactivity disorder and early-onset conduct disorder: Biological, environmental, and developmental mechanisms. *Clinical Psychology: Science and Practice, 17,* 327–336. http://dx.doi.org/10.1111/j.1468-2850.2010.01224.x

Biederman, J., Faraone, S., Milberger, S., Curtis, S., Chen, L., Marrs, A., . . . Spencer, T. (1996). Predictors of persistence and remission of ADHD into adolescence: Results from a four-year prospective follow-up study. *Journal of the American Academy of Child & Adolescent Psychiatry, 35,* 343–351. http://dx.doi.org/10.1097/00004583-199603000-00016

Biederman, J., Faraone, S. V., Milberger, S., Jetton, J. G., Chen, L., Mick, E., . . . Russell, R. L. (1996). Is childhood oppositional defiant disorder a precursor to adolescent conduct disorder? Findings from a four-year follow-up study of children with ADHD. *Journal of the American Academy of Child & Adolescent Psychiatry, 35,* 1193–1204. http://dx.doi.org/10.1097/00004583-199609000-00017

Bloom, B., Jones, L. I., & Freeman, G. (2013). Summary health statistics for U.S. children: National Health Interview Survey. *National Center for Health Statistics*. Vital Health Statistics, *2012,* 10(258).

Bor, W., Sanders, M. R., & Markie-Dadds, C. (2002). The effects of the Triple P-Positive Parenting Program on preschool children with co-occurring disruptive behavior and attentional/hyperactive difficulties. *Journal of Abnormal Child Psychology, 30,* 571–587. http://dx.doi.org/10.1023/A:1020807613155

Burke, J. D., Loeber, R., Lahey, B. B., & Rathouz, P. J. (2005). Developmental transitions among affective and behavioral disorders in adoles-

cent boys. *Journal of Child Psychology and Psychiatry, 46,* 1200–1210. http://dx.doi.org/10.1111/j.1469-7610.2005.00422.x

Burns, G. L., & Walsh, J. A. (2002). The influence of ADHD-hyperactivity/impulsivity symptoms on the development of oppositional defiant disorder symptoms in a 2-year longitudinal study. *Journal of Abnormal Child Psychology, 30,* 245–256. http://dx.doi.org/10.1023/A:1015102812958

Burt, S. A., Krueger, R. F., McGue, M., & Iacono, W. G. (2001). Sources of covariation among attention-deficit/hyperactivity disorder, oppositional defiant disorder, and conduct disorder: The importance of shared environment. *Journal of Abnormal Psychology, 110,* 516–525. http://dx.doi.org/10.1037/0021-843X.110.4.516

Campbell, S. B. (2002). *Behavior problems in preschool children: Clinical and developmental issues* (2nd ed.). New York, NY: Guilford Press.

Campbell, S. B., Breaux, A. M., Ewing, L. J., & Szumowski, E. K. (1986). Correlates and predictors of hyperactivity and aggression: A longitudinal study of parent-referred problem preschoolers. *Journal of Abnormal Child Psychology, 14,* 217–234. http://dx.doi.org/10.1007/BF00915442

Caspi, A., Langley, K., Milne, B., Moffitt, T. E., O'Donovan, M., Owen, M. J., . . . Thapar, A. (2008). A replicated molecular genetic basis for subtyping antisocial behavior in children with attention-deficit/hyperactivity disorder. *Archives of General Psychiatry, 65,* 203–210. http://dx.doi.org/10.1001/archgenpsychiatry.2007.24

Comings, D. E., Gade-Andavolu, R., Gonzalez, N., Wu, S., Muhleman, D., Blake, H., . . . MacMurray, J. P. (2000). Multivariate analysis of associations of 42 genes in ADHD, ODD and conduct disorder. *Clinical Genetics, 58,* 31–40. http://dx.doi.org/10.1034/j.1399-0004.2000.580106.x

Denham, S. A. (2006). Emotional competence: Implications for social functioning. In J. Luby (Ed.), *Handbook of preschool mental health: Development, disorders, and treatment* (pp. 23–44). New York, NY: Guilford Press.

Diamantopoulou, S., Verhulst, F. C., & van der Ende, J. (2010). Testing developmental pathways to antisocial personality problems. *Journal of Abnormal Child Psychology, 38,* 91–103. http://dx.doi.org/10.1007/s10802-009-9348-7

Doerr, M., & Teng, K. (2012). Family history: Still relevant in the genomics era. *Cleveland Clinic Journal of Medicine, 79,* 331–336. http://dx.doi.org/10.3949/ccjm.79a.11065

Eaves, L., Rutter, M., Silberg, J. L., Shillady, L., Maes, H., & Pickles, A. (2000). Genetic and environmental causes of covariation in interview assessments of disruptive behavior in child and adolescent twins. *Behavior Genetics, 30,* 321–334. http://dx.doi.org/10.1023/A:1026553518272

Enders, C. K., & Bandalos, D. L. (2001). The relative performance of full information maximum likelihood estimation for missing data in structural equation models. *Structural Equation Modeling, 8,* 430–457. http://dx.doi.org/10.1207/S15328007SEM0803_5

Faraone, S. V., Biederman, J., & Monuteaux, M. C. (2000). Attention-deficit disorder and conduct disorder in girls: Evidence for a familial subtype. *Biological Psychiatry, 48,* 21–29. Retrieved from http://www.ncbi.nlm.nih.gov/pubmed/10913504. http://dx.doi.org/10.1016/S0006-3223(00)00230-4

Frick, P. J., Lahey, B. B., Christ, M. A. G., Loeber, R., & Green, S. (1991). History of childhood behavior problems in biological relatives of boys with attention-deficit hyperactivity disorder and conduct disorder. *Journal of Clinical Child Psychology, 20,* 445–451. http://dx.doi.org/10.1207/s15374424jccp2004_14

Frick, P. J., Lahey, B. B., Loeber, R., Stouthamer-Loeber, M., Christ, M. A. G., & Hanson, K. (1992). Familial risk factors to oppositional defiant disorder and conduct disorder: Parental psychopathology and maternal parenting. *Journal of Consulting and Clinical Psychology, 60,* 49–55. http://dx.doi.org/10.1037/0022-006X.60.1.49

Galéra, C., Côté, S. M., Bouvard, M. P., Pingault, J.-B., Melchior, M., Michel, G., . . . Tremblay, R. E. (2011). Early risk factors for hyperactivity-impulsivity and inattention trajectories from age 17 months to 8 years. *Archives of General Psychiatry, 68,* 1267–1275. http://dx.doi.org/10.1001/archgenpsychiatry.2011.138

Gershon, J. (2002). A meta-analytic review of gender differences in ADHD. *Journal of Attention Disorders, 5,* 143–154. http://dx.doi.org/10.1177/108705470200500302

Goldstein, L. H., Harvey, E. A., & Friedman-Weieneth, J. L. (2007). Examining subtypes of behavior problems among 3-year-old children, Part III: Investigating differences in parenting practices and parenting stress. *Journal of Abnormal Child Psychology, 35,* 125–136. http://dx.doi.org/10.1007/s10802-006-9047-6

Goldstein, L. H., Harvey, E. A., Friedman-Weieneth, J. L., Pierce, C., Tellert, A., & Sippel, J. C. (2007). Examining subtypes of behavior problems among 3-year-old children, Part II: Investigating differences in parent psychopathology, couple conflict, and other family stressors. *Journal of Abnormal Child Psychology, 35,* 111–123. http://dx.doi.org/10.1007/s10802-006-9088-x

Harvey, E. A., Friedman-Weieneth, J. L., Goldstein, L. H., & Sherman, A. H. (2007). Examining subtypes of behavior problems among 3-year-old children, Part I: Investigating validity of subtypes and biological risk-factors. *Journal of Abnormal Child Psychology, 35,* 97–110. http://dx.doi.org/10.1007/s10802-006-9087-y

Harvey, E. A., Lugo-Candelas, C. I., & Breaux, R. P. (2015). Longitudinal changes in individual symptoms across the preschool years in children with ADHD. *Journal of Clinical Child and Adolescent Psychology, 44,* 580–594. http://dx.doi.org/10.1080/15374416.2014.886253

Harvey, E. A., Metcalfe, L. A., Herbert, S. D., & Fanton, J. H. (2011). The role of family experiences and ADHD in the early development of oppositional defiant disorder. *Journal of Consulting and Clinical Psychology, 79,* 784–795. http://dx.doi.org/10.1037/a0025672

Harvey, E. A., Youngwirth, S. D., Thakar, D. A., & Errazuriz, P. A. (2009). Predicting attention-deficit/hyperactivity disorder and oppositional defiant disorder from preschool diagnostic assessments. *Journal of Consulting and Clinical Psychology, 77,* 349–354. http://dx.doi.org/10.1037/a0014638

Herbert, S. D., Harvey, E. A., Roberts, J. L., Wichowski, K., & Lugo-Candelas, C. I. (2013). A randomized controlled trial of a parent training and emotion socialization program for families of hyperactive preschool-aged children. *Behavior Therapy, 44,* 302–316. http://dx.doi.org/10.1016/j.beth.2012.10.004

Hoza, B., Mrug, S., Gerdes, A. C., Hinshaw, S. P., Bukowski, W. M., Gold, J. A., . . . Arnold, L. E. (2005). What aspects of peer relationships are impaired in children with attention-deficit/hyperactivity disorder? *Journal of Consulting and Clinical Psychology, 73,* 411–423. http://dx.doi.org/10.1037/0022-006X.73.3.411

Jackson, D. A., & King, A. R. (2004). Gender differences in the effects of oppositional behavior on teacher ratings of ADHD symptoms. *Journal of Abnormal Child Psychology, 32,* 215–224. http://dx.doi.org/10.1023/B:JACP.0000019772.71251.ff

Johnston, C., & Jassy, J. S. (2007). Attention-deficit/hyperactivity disorder and oppositional/conduct problems: Links to parent-child interactions. *Journal of the Canadian Academy of Child and Adolescent Psychiatry, 16,* 74–79.

Johnston, C., & Mash, E. J. (2001). Families of children with attention-deficit/hyperactivity disorder: Review and recommendations for future research. *Clinical Child and Family Psychology Review, 4,* 183–207. Retrieved from http://www.ncbi.nlm.nih.gov/pubmed/11783738. http://dx.doi.org/10.1023/A:1017592030434

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This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.

Kaiser, N. M., McBurnett, K., & Pfiffner, L. J. (2011). Child ADHD severity and positive and negative parenting as predictors of child social functioning: Evaluation of three theoretical models. *Journal of Attention Disorders, 15,* 193–203. http://dx.doi.org/10.1177/1087054709356171

Keown, L. J., & Woodward, L. J. (2006). Preschool boys with pervasive hyperactivity: Early peer functioning and mother-child relationship influences. *Social Development, 15,* 23–45. http://dx.doi.org/10.1111/j.1467-9507.2006.00328.x

Kuny, A. V., Althoff, R. R., Copeland, W., Bartels, M., Van Beijsterveldt, C. E. M., Baer, J., & Hudziak, J. J. (2013). Separating the domains of oppositional behavior: Comparing latent models of the Conner's' oppositional subscale. *Journal of the American Academy of Child & Adolescent Psychiatry, 52,* 172–183.e8. http://dx.doi.org/10.1016/j.jaac.2012.10.005

Lahey, B. B., McBurnett, K., & Loeber, R. (2000). Are attention-deficit/hyperactivity disorder and oppositional defiant disorder developmental precursors to conduct disorder? In A. J. Sameroff, M. Lewis, & S. Miller (Eds.), *Handbook of developmental psychopathology* (2nd ed., pp. 431–446). Dordrecht, Netherlands: Kluwer Academic Publishers. http://dx.doi.org/10.1007/978-1-4615-4163-9_23

Lahey, B. B., Pelham, W. E., Loney, J., Kipp, H., Ehrhardt, A., Lee, S. S., . . . Massetti, G. (2004). Three-year predictive validity of *DSM–IV* attention deficit hyperactivity disorder in children diagnosed at 4–6 years of age. *The American Journal of Psychiatry, 161,* 2014–2020. http://dx.doi.org/10.1176/appi.ajp.161.11.2014

Lahey, B. B., Van Hulle, C. A., Rathouz, P. J., Rodgers, J. L., D'Onofrio, B. M., & Waldman, I. D. (2009). Are oppositional-defiant and hyperactive-inattentive symptoms developmental precursors to conduct problems in late childhood? Genetic and environmental links. *Journal of Abnormal Child Psychology, 37,* 45–58. http://dx.doi.org/10.1007/s10802-008-9257-1

Larson, K., Russ, S. A., Kahn, R. S., & Halfon, N. (2011). Patterns of comorbidity, functioning, and service use for US children with ADHD, 2007. *Pediatrics, 127,* 462–470. http://dx.doi.org/10.1542/peds.2010-0165

Lavigne, J. V., Cicchetti, C., Gibbons, R. D., Binns, H. J., Larsen, L., & DeVito, C. (2001). Oppositional defiant disorder with onset in preschool years: Longitudinal stability and pathways to other disorders. *Journal of the American Academy of Child & Adolescent Psychiatry, 40,* 1393–1400. http://dx.doi.org/10.1097/00004583-200112000-00009

Lavigne, J. V., Herzing, L. B. K., Cook, E. H., Lebailly, S. A., Gouze, K. R., Hopkins, J., & Bryant, F. B. (2013). Gene × environment effects of serotonin transporter, dopamine receptor D4, and monoamine oxidase A genes with contextual and parenting risk factors on symptoms of oppositional defiant disorder, anxiety, and depression in a community sample of 4-year-old children. *Development and Psychopathology, 25,* 555–575. http://dx.doi.org/10.1017/S0954579412001241

Lavigne, J. V., Lebailly, S. A., Hopkins, J., Gouze, K. R., & Binns, H. J. (2009). The prevalence of ADHD, ODD, depression, and anxiety in a community sample of 4-year-olds. *Journal of Clinical Child and Adolescent Psychology, 38,* 315–328. http://dx.doi.org/10.1080/15374410902851382

Miller-Johnson, S., Coie, J. D., Maumary-Gremaud, A., & Bierman, K., & the Conduct Problems Prevention Research Group. (2002). Peer rejection and aggression and early starter models of conduct disorder. *Journal of Abnormal Child Psychology, 30,* 217–230. http://dx.doi.org/10.1023/A:1015198612049

Muthén, L. K., & Muthén, B. O. (1998–2010). *Mplus User's Guide* (6th Ed.). Los Angeles, CA: Authors.

Nock, M. K., Kazdin, A. E., Hiripi, E., & Kessler, R. C. (2007). Lifetime prevalence, correlates, and persistence of oppositional defiant disorder: Results from the National Comorbidity Survey Replication. *Journal of Child Psychology and Psychiatry, 48,* 703–713. http://dx.doi.org/10.1111/j.1469-7610.2007.01733.x

Pardini, D. A., & Fite, P. J. (2010). Symptoms of conduct disorder, oppositional defiant disorder, attention-deficit/hyperactivity disorder, and callous-unemotional traits as unique predictors of psychosocial maladjustment in boys: Advancing an evidence base for DSM-V. *Journal of the American Academy of Child & Adolescent Psychiatry, 49,* 1134–1144.

Pauli-Pott, U., Dalir, S., Mingebach, T., Roller, A., & Becker, K. (2013). Do different ADHD-related etiological risks involve specific neuropsychological pathways? An analysis of mediation processes by inhibitory control and delay aversion. *Journal of Child Psychology and Psychiatry, 54,* 800–809. http://dx.doi.org/10.1111/jcpp.12059

Petty, C. R., Monuteaux, M. C., Mick, E., Hughes, S., Small, J., Faraone, S. V., & Biederman, J. (2009). Parsing the familiality of oppositional defiant disorder from that of conduct disorder: A familial risk analysis. *Journal of Psychiatric Research, 43,* 345–352. http://dx.doi.org/10.1016/j.jpsychires.2008.03.010

Pisterman, S., Firestone, P., McGrath, P., Goodman, J. T., Webster, I., Mallory, R., & Goffin, B. (1992). The role of parent training in treatment of preschoolers with ADDH. *American Journal of Orthopsychiatry, 62,* 397–408. http://dx.doi.org/10.1037/h0079356

Reynolds, C. R., & Kamphaus, R. W. (1992). *Behavior Assessment System for Children*. Circle Pines, MN: American Guidance Service.

Rhee, S. H., Willcutt, E. G., Hartman, C. A., Pennington, B. F., & DeFries, J. C. (2008). Test of alternative hypotheses explaining the comorbidity between attention-deficit/hyperactivity disorder and conduct disorder. *Journal of Abnormal Child Psychology, 36,* 29–40. http://dx.doi.org/10.1007/s10802-007-9157-9

Riddle, M. A., Yershova, K., Lazzaretto, D., Paykina, N., Yenokyan, G., Greenhill, L., . . . Posner, K. (2013). The preschool attention-deficit/hyperactivity disorder treatment study (PATS) 6-year follow-up. *Journal of the American Academy of Child & Adolescent Psychiatry, 52,* 264–278.e2. http://dx.doi.org/10.1016/j.jaac.2012.12.007

Shaffer, D., Fisher, P., Lucas, C. P., Dulcan, M. K., & Schwab-Stone, M. E. (2000). NIMH Diagnostic Interview Schedule for Children Version IV (NIMH DISC-IV): Description, differences from previous versions, and reliability of some common diagnoses. *Journal of the American Academy of Child & Adolescent Psychiatry, 39,* 28–38. http://dx.doi.org/10.1097/00004583-200001000-00014

Silberg, J., Rutter, M., Meyer, J., Maes, H., Hewitt, J., Simonoff, E., . . . Eaves, L. (1996). Genetic and environmental influences on the covariation between hyperactivity and conduct disturbance in juvenile twins. *Journal of Child Psychology and Psychiatry, 37,* 803–816. http://dx.doi.org/10.1111/j.1469-7610.1996.tb01476.x

Smalley, S. L., McGough, J. J., Del'Homme, M., NewDelman, J., Gordon, E., Kim, T., . . . McCracken, J. T. (2000). Familial clustering of symptoms and disruptive behaviors in multiplex families with attention-deficit/hyperactivity disorder. *Journal of the American Academy of Child & Adolescent Psychiatry, 39,* 1135–1143. http://dx.doi.org/10.1097/00004583-200009000-00013

Sonuga-Barke, E. J. S., Oades, R. D., Psychogiou, L., Chen, W., Franke, B., Buitelaar, J., . . . Faraone, S. V. (2009). Dopamine and serotonin transporter genotypes moderate sensitivity to maternal expressed emotion: The case of conduct and emotional problems in attention deficit/hyperactivity disorder. *Journal of Child Psychology and Psychiatry, 50,* 1052–1063. http://dx.doi.org/10.1111/j.1469-7610.2009.02095.x

Speltz, M. L., McClellan, J., DeKlyen, M., & Jones, K. (1999). Preschool boys with oppositional defiant disorder: Clinical presentation and diagnostic change. *Journal of the American Academy of Child & Adolescent Psychiatry, 38,* 838–845. http://dx.doi.org/10.1097/00004583-199907000-00013

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Stringaris, A., & Goodman, R. (2009). Longitudinal outcome of youth oppositionality: Irritable, headstrong, and hurtful behaviors have distinctive predictions. *Journal of the American Academy of Child & Adolescent Psychiatry, 48,* 404–412. http://dx.doi.org/10.1097/CHI.0b013e3181984f30

Tandon, M., Si, X., & Luby, J. (2011). Preschool onset attention-deficit/hyperactivity disorder: Course and predictors of stability over 24 months. *Journal of Child and Adolescent Psychopharmacology, 21,* 321–330. http://dx.doi.org/10.1089/cap.2010.0045

Tuvblad, C., Zheng, M., Raine, A., & Baker, L. A. (2009). A common genetic factor explains the covariation among ADHD ODD and CD symptoms in 9–10 year old boys and girls. *Journal of Abnormal Child Psychology, 37,* 153–167. http://dx.doi.org/10.1007/s10802-008-9278-9

Wåhlstedt, C., Thorell, L. B., & Bohlin, G. (2008). ADHD symptoms and executive function impairment: Early predictors of later behavioral problems. *Developmental Neuropsychology, 33,* 160–178. http://dx.doi.org/10.1080/87565640701884253

Waschbusch, D. A. (2002). A meta-analytic examination of comorbid hyperactive-impulsive-attention problems and conduct problems. *Psychological Bulletin, 128,* 118–150. http://dx.doi.org/10.1037/0033-2909.128.1.118

Yeh, M. T., Coccaro, E. F., & Jacobson, K. C. (2010). Multivariate behavior genetic analyses of aggressive behavior subtypes. *Behavior Genetics, 40,* 603–617. http://dx.doi.org/10.1007/s10519-010-9363-z

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